

AI3613 Database System Concepts Homework 1

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Problem 1

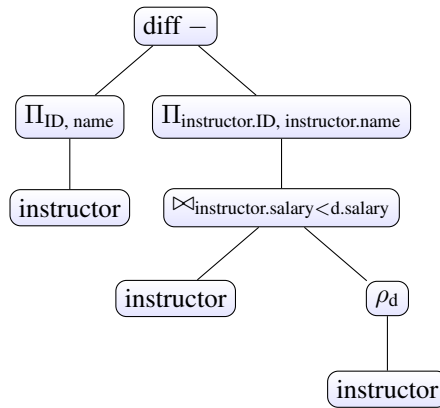
1. For the first sub-problem, we provide the result first as it is given in class as

$$\Pi_{\text{salary}}(\text{instructor}) - \Pi_{\text{instructor.salary}}(\text{instructor} \bowtie_{\text{instructor.salary} < \text{d.salary}} \rho_d(\text{instructor})),$$

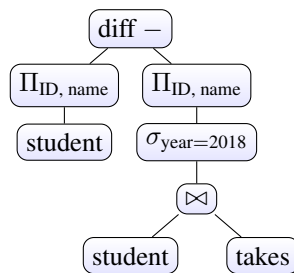
and simply change it to

$$\Pi_{\text{ID, name}}(\text{instructor}) - \Pi_{\text{instructor.ID, instructor.name}}(\text{instructor} \bowtie_{\text{instructor.salary} < \text{d.salary}} \rho_d(\text{instructor})).$$

Expression Tree:



2. Expression Tree:

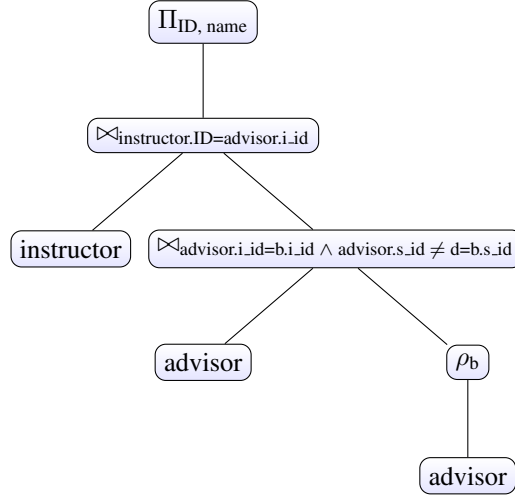


The expression is

$$\Pi_{\text{ID, name}}(\text{student}) - \Pi_{\text{ID, name}}(\sigma_{\text{year}=2018}(\text{student} \bowtie \text{takes}))$$

3. Concat two advisor tables first, then select same rows with same instructor and different students.

Expression Tree:



The expression is

$$\Pi_{ID, name}(\text{instructor} \bowtie_{\text{instructor.ID=advisor.i.id}} (\text{advisor} \bowtie_{\text{advisor.i.id=b.i.id} \wedge \text{advisor.s.id} \neq \text{d=b.s.id}} \rho_b(\text{advisor})))$$

Problem 2

We suppose that the operations is with DISTICT, as mentioned in calss.

From all the possible ts , we exlude the ts that do not satisfy the condition. We cam generate all possible ts by $\Pi_{A_1, A_2 \dots, A_n}(R) \times S$. The desired ts are in R , and thus the rest is $\Pi_{A_1, A_2 \dots, A_n}(R) \times S - R$.

We simply get the desired results by a difference operation.

$$\Pi_{A_1, A_2 \dots, A_n}(R) - \Pi_{A_1, A_2 \dots, A_n}(\Pi_{A_1, A_2 \dots, A_n}(R) \times S - R)$$

Problem 3

Let's congratulate the TAs for the second prize in the graph pattern matching competition lol.